

C Programming - Assignment-06

Q.1 Write command line program to check if a string is palindrome or not.

Concepts Used

- Command Line Arguments
- Strings
- String Length (`strlen`)
- Looping (`for`)
- Conditional Statements (`if`)
- Character Comparison

Step-by-Step Approach

1. Accept string using command line argument.
2. Find length of string using `strlen()`.
3. Compare characters from both ends.
4. If all characters match, string is palindrome.
5. Otherwise, string is not palindrome.
6. Display the result.

Test Cases

Test Case	Input	Output
1	ABA	ABA is palindrome
2	SGGS	SGGS is palindrome
3	xcvDbmMbDvcx	xcvDbmMbDvcx is not a palindrome
4	ASMpMSA	ASMpMSA is a palindrome
5	KMCVDS DVCNK	KMCVDS DVCNK is not a palindrome

Q.2 Write a command line program to convert all vowels in a given string to uppercase.

Concepts Used

- Command Line Arguments
- Strings
- Looping (`for`)
- Conditional Statements (`if`)
- Character Manipulation
- `toupper()` Function

Step-by-Step Approach

1. Accept string using command line argument.
2. Traverse each character of the string.
3. Check whether the character is a vowel.
4. Convert vowels to uppercase.
5. Display the modified string.

Test Cases

Test Case	Input	Output
1	hello	hELLO
2	computer	cOmpUtEr
3	programming	prOgrAmmlng
4	structure	strUctUrE

Q.3 Write a command line program to check if a given string is present in another string.

Concepts Used

- Command Line Arguments
- Strings
- String Functions (`strstr`)
- Conditional Statements (`if`)

Step-by-Step Approach

1. Accept two strings using command line arguments.
2. Search first string in second string.
3. Use `strstr()` function.
4. If found, print "Yes".
5. Otherwise, print "No".

Test Cases

Test Case	Input	Output
1	ing Programming	Yes
2	pac compact	Yes
3	Engineer TextileAndEngineeringInstitute	Yes
4	Istitute TextileAndEngineeringInstitute	No